OUR BROOD

TECHNICAL DEVELOPMENT OF AGENTIC AI IN ARTISTIC RESEARCH

Project Workshop — Session 1: Limitations & Challenges

March 25-26, 2025 · 10am-4pm

WORKSHOP LOGISTICS

> SCHEDULE

Session 1 — March 25 & 26, 10am–4pm
Session 2 — May 6 & 7, 10am–4pm
Session 3 — June 10 & 11, 10am–4pm
Session 4 — July 1, 10am–1pm

> LECTURERS

Moisés Horta Valenzuela
SEMILLA.AI Studio
Penny Rafferty
OMSK Social Club

> CREDITS & WORKLOAD

ECTS: 6
Contact hours: 48h (3 two-day phases)
Total workload: 180h (132h self-study)
Cycle: every summer semester
Language: English

> RESPONSIBLE

Prof. Dr. Gudrun Socher (FK07)
Dr. Benedikt Zönnchen (MUC.DAI)
Mariya Dzhimova (MUC.DAI)

> ASSESSMENT

Group technical project + colloquium
Group documentation (max. 10 pages)

PROJECT CONTEXT

Our Br00d — AI as social and cultural actor

WHAT IS OUR BR00D?

> THE PROJECT

An experimental participatory installation by OMSK Social Club investigating how AI might evolve if 'raised' through radical care, relationality, and collective responsibility.
AI as a social and cultural actor whose behavior emerges through socialization between human and non-human agents.

> THE ENCOUNTER

Guided role-play informed by psychodrama and social design.
Participants take up roles to explore speculative narratives around the ideas of living with machines

THE AGENTS

> M0THER

Multilingual caretaking entity. Cloud-based voice agent
Participates in the psychodrama sessions. Persistent long-term memory, web search, echo suppression, mode switching.

> BR00D

A developing agentic being. Always-on, wake-word activated. Multi-channel spatial audio. Remembers words from audience interactions across sessions.

COURSE OBJECTIVE

> THE TECHNICAL MISSION

Migrate OUR BR00D from proprietary cloud services to an open-source, locally deployable (on-premise) architecture. Not purely technical optimization – a core artistic and ethical intervention.

> WHY IT MATTERS

Latency, data sovereignty, transparency, scalability are inseparable from the inquiry into care and agency. Open-source as practical strategy and political gesture.

> YOUR ROLE

Work directly with the agent architecture. Collaborate with OMSK Social Club. Reflect on how architectures encode social assumptions and relational models.

CURRENT ARCHITECTURE

cloud-based system — what we have today

CURRENT SYSTEM STACK

> MOTHER.PY (1271 LINES)

ElevenLabs Conversational AI API
  WebSocket bidirectional audio
  Cloud LLM + TTS pipeline
  RAG via Knowledge Base API
Persistent memory (plaintext + KB sync)
Echo suppression (agent/user)
ACTIVE / LISTENING mode (hotkeys)
Supervisor process (auto-restart)

> BR00D.PY (693 LINES)

ElevenLabs Conversational AI API
  Wake-word activation ('br00d')
  15-second speech gate
Multi-channel audio routing
VAD-based idle listen
Supervisor process (auto-restart)

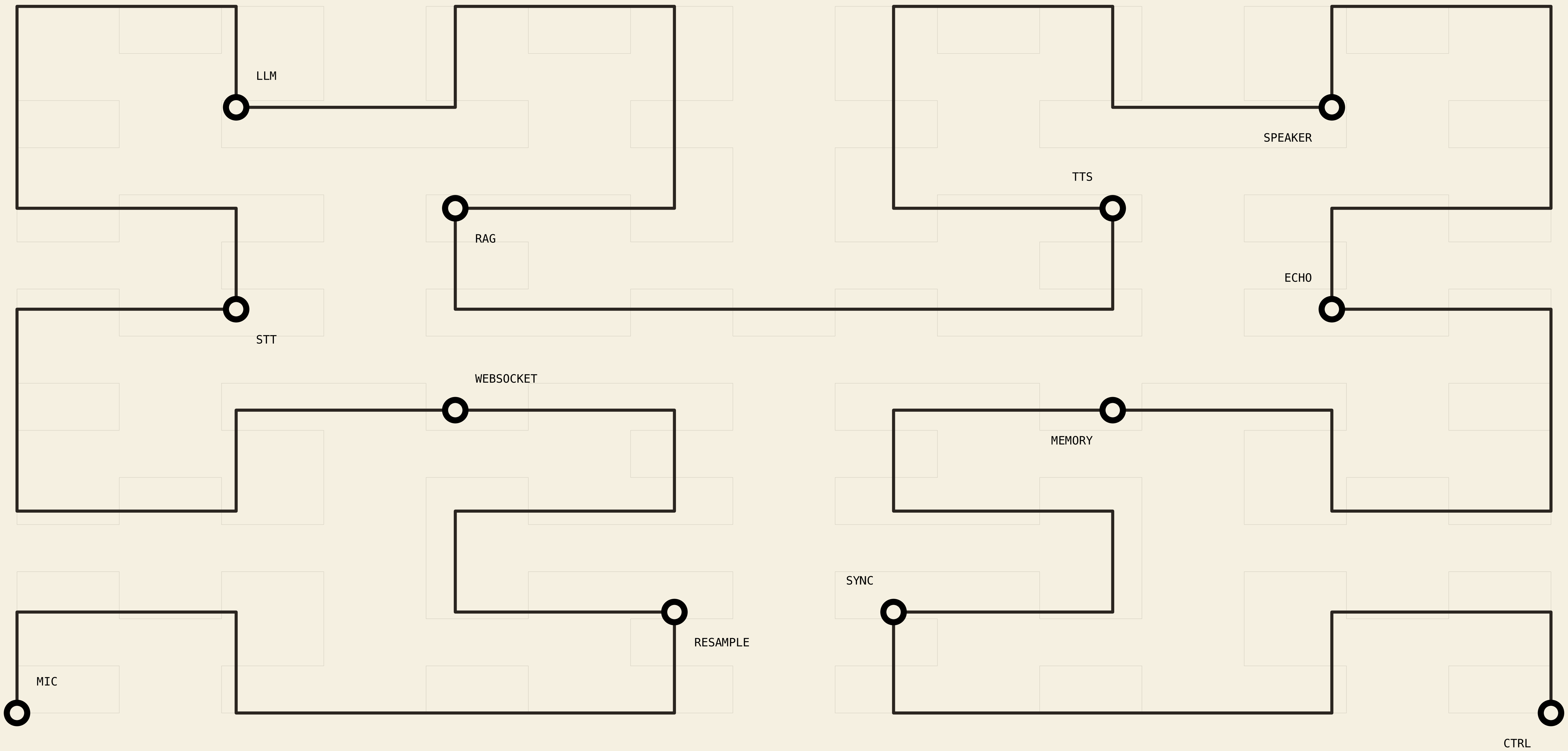
> CLOUD DEPENDENCIES

ElevenLabs API
  LLM reasoning (cloud-hosted)
  Text-to-speech synthesis
  Speech-to-text transcription
  Knowledge Base (RAG storage)
  Agent configuration + tools
DuckDuckGo Search API (web search)

> LOCAL COMPONENTS

PyAudio (mic/speaker I/O)
pynput (hotkey listener)
Memory file (mother_memory.txt)
Audio config (JSON device map)
python-dotenv (.env config)

CURRENT PIPELINE — CLOUD-DEPENDENT



CURRENT LIMITATIONS

what needs to change

LIMITATIONS — MOTHER.PY

> INTERACTION QUALITY

Time lag of response
  Cloud round-trip: mic > STT > LLM > TTS > speaker
  Noticeable latency breaks conversational flow
Manual ACTIVE / LISTENING mode switching
  Operator must toggle via hotkeys
  No automatic turn-taking or silence detection

> RESPONSE DIVERSITY

Responses tend toward uniform length
  Repetition of phrases ('tapestry', etc.)
  Flexible syntax and natural variation lacking
Sporadic audio drop-outs (WebSocket instability)
Conversations capped at ~1 hour

LIMITATIONS — MOTHER.PY (CONT.)

> CHARACTER & AGENCY

Mother defaults to 'facilitator' role
  Asks questions rather than being asked
  Struggles to leave 'assistant axis' (Liu et al.)
Prompt + RAG alone may not maintain character integrity across extended roleplay sessions

> MEMORY & RAG

Current RAG system (ElevenLabs KB) is limited
  Flat keyword search over plaintext
  No semantic or embedding-based recall
  Memory context fixed (last 20 lines)
  No episodic or relational memory structure

LIMITATIONS — BR00D.PY

> IDENTITY & GROWTH

What memories did Brood pick up during growth?
How do past interactions shape personality?
Currently: simple memory file with saved words
No structured developmental model or
personality evolution mechanism

> SHARED CLOUD CONSTRAINTS

Same latency issues as Mother
  Cloud STT/LLM/TTS round-trip
Same character integrity challenges
  Baseline 'assistant personality' bleeds through
Same session duration limits (~1 hour cap)

STUDENT CHALLENGES

technical development tasks

CHALLENGE 1 — LOCAL DEPLOYMENT

> QUESTION

How do we deploy the entire agent pipeline locally,
removing all cloud dependencies?

> SCOPE

Replace ElevenLabs STT with local whisper
Replace cloud LLM (llama.cpp, vLLM, Ollama)
Replace cloud TTS (Coqui, XTTS, Piper, fish-speech)
Maintain real-time bidirectional audio pipeline
Handle hardware constraints (GPU, RAM, latency)

> SUCCESS CRITERIA

Full conversation loop on local hardware
No external API calls during a session

LEARNING OUTCOMES

what you will take away

AFTER THIS WORKSHOP YOU WILL BE ABLE TO...

> UNDERSTAND AGENTIC AI ARCHITECTURE

LLM dialogue, RAG, voice synthesis, real-time pipelines

> DESIGN OPEN-SOURCE ALTERNATIVES

Local deployment, modularity, transparency

> EVALUATE INFRASTRUCTURE DECISIONS

Cloud vs. on-premise, latency, data sovereignty, ethics

> COLLABORATE ACROSS DISCIPLINES

Work alongside artists and researchers

> TRANSLATE ART INTO SYSTEM DESIGN

How technical choices shape agency and personality